

Towards Virtual Teaching Hospitals for Advanced Surgical Training

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Figure 1: Rooms in a virtual teaching hospital. Left: Training room for surgical planning that contains four training stations, including virtual organ library, information board, *DICOM* board, and virtual resection planning. Right: a virtual operating room for laparoscopic surgery training.

ABSTRACT

Existing virtual training environments in medicine usually focus only on the training of a specific medical skill and are conducted in a single virtual room. However, medical challenges are often in the context of the planning procedure and operating room intervention. Therefore, it is crucial to provide a training environment that tackles these issues. In this paper, we present a prototype of surgical department within a virtual teaching hospital. It supports multiple users, photo-realistic avatars, and training scenarios ranging from planning procedures to laparoscopic surgery in the virtual operating room. Medical data for each virtual patient is synchronized across the different rooms, allowing users to view, analyze and train on key decision points from diagnosis to surgical incision for each patient. The environment was evaluated in a pilot study with five surgical residents and one medical student. The experts assessed the environment as an essential tool to evaluate and improve surgical planning as well as the training during critical situations of the surgical procedures. We conclude that the development of the virtual teaching hospital components described here has the potential to be a basis for future generations of medical training simulators.

Index Terms: I.3.7 [COMPUTER GRAPHICS]: Three-Dimensional Graphics and Realism—Virtual reality; K.3.1 [COMPUTERS AND EDUCATION]: Computer Uses in Education—Collaborative learning

1 INTRODUCTION

Surgical planning and training could improve accessibility and flexibility through interactions with segmented 3D models. However, it is essential to provide all related information, including patient history, medical image data, and virtual resection planning [1–3].

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Therefore, a combination of providing surgical planning information with an integrated operating room in a virtual teaching hospital, could enable more advanced surgical training.

Apart from the solutions which are mostly designed as single-user applications, collaborative virtual reality (VR), where multiple users share a virtual environment, offers even great potential for team collaboration [4, 5]. It allows a group of users including surgical trainees and experienced surgeons, to connect and train within a shared virtual environment. Additionally, collaborative VR can improve the communication and teamwork, which eventually can support interprofessional team training [6, 7].

In this work, we present an advanced surgical training environment within a virtual teaching hospital. The work builds on our previous research in the field of virtual surgery planning and training. It also supports photo-realistic avatars, where users can generate their representation models from 2D photos. For surgical planning, we developed a virtual room that supports various collaborative aspects for patient data exploration and planning. It includes an enhanced version of liver anatomy education proposed by Schott et al. [8] and a VR-based technique for virtual resection planning [9]. In addition, we integrated a laparoscopic simulation with spatial input devices and training scenarios for interprofessional team training into the virtual operating room [6, 7]. This allows the users to perform the training with different user roles, including surgeons, camera assistants, and anesthesiologists. Thus, they can train for critical and distracting events, where communication between each profession is necessary.

2 RELATED WORK

Different conceptual aspects, including liver anatomy, tumor localization, diagnosis and patient information, are essential for liver surgery education and training [10]. Nobuoka et al. [11] proposed a video-based system for liver anatomy education. Their multi-layer liver anatomy atlas was developed by filming a real dissection. However, the recorded videos can be watched by students passively. Birr et al. [12] presented the *LiverAnatomyExplorer*, which is a web-based application that combines 2D images, 3D models, surgical videos, and assessment tools. Kurul et al. [13] indicates that anatomy

training within an immersive VR environment leads to higher test scores. Additionally, study results from Sankaran et al. [14] show that realizing medical education within VR environment could accelerate the clinical education. Weyhe et al. [15] proposed a virtual 3D anatomy atlas in a VR environment. The results of their study show that their VR system led to fast learning success and a higher satisfaction compared to conventional learning. Nonetheless, in contrast to our work, the previously mentioned systems do not allow collaboration of multiple users in the same shared virtual environment.

2.1 Surgical Education and Training

Nigel John [16] described the use of web-based technologies for medical education and training. Furcea et al. [17] described a web-based e-learning platform that combines elements for pre-planning of the operation with the training of laparoscopic liver surgery. Mönch et al. [18] presented the *LiverSurgeryTrainer*, a desktop application where surgeons could train their decisions with individual resections and benefit from comparisons to expert suggestions for a number of cases that differ in complexity. A similar tool with focus on pre-operative planning of complex liver surgeries was presented by Reitingier et al. [19]. Their *LiverPlanner* was realized within a semi-immersive environment with a large-screen stereoscopic projection system. As an input device, a combination of a tablet and a controller is used. In contrast to our system, these aim to prepare and plan individual surgical case and not to train interventions.

2.2 Collaborative VR for Surgical Training

Collaborative VR emerged as an essential topic for research [20]. In general, there is a positive educational effect if multiple users share one virtual environment [21, 22]. A group of users with different professions can explore complex anatomical structures together and train for different interventions. An effective user constellation is a one-on-one setting, where a teacher supports one student and guides them within a specific scenario [23]. Paiva et al. [24] presented a VR-based simulator for team training and put emphasis on the distinction between individual tasks, collaborative tasks, and simulated surgical procedures. Prasolova et al. [25] presented a virtual learning environment for interprofessional team communication and collaboration, including the role of the anesthesiologist. They could show that their training lead to more engaged and motivated students, who understood the importance of clear communication. A way to improve communication in anesthesia was presented by Cordar et al. [26]. They used agents on large monitors in a surgery setting to train anesthesia residents and observed significant communication difference in the residents, depending on the behavior of the agents. Furthermore, Brunges et al. [27] presented a similar mixed reality technology for interdisciplinary teams. Their results indicated that ineffective preoperative team member communication can result in serious patient harm. While Cordar and Bruges et al. focused on monitor-based solutions, our proposed system allows multi-user to connect and perform the training in the immersive environment.

3 DESIGN OF A VIRTUAL TEACHING HOSPITAL

In the following, we describe the proposed collaborative VR environment that can serve as an exploration, planning, and training environment for advanced surgical training. There are three main rooms in this environment: a lobby, a surgical planning room, and a virtual operating room (see Fig. 2). Additionally, the system architecture and evaluation are described in this section.

3.1 Lobby

After each group member chooses their role in the user management system, a leader can press the start button to allow other members to join in the collaborative VR environment. Thereafter, they will connect and form as a group in the lobby. In this room, they can

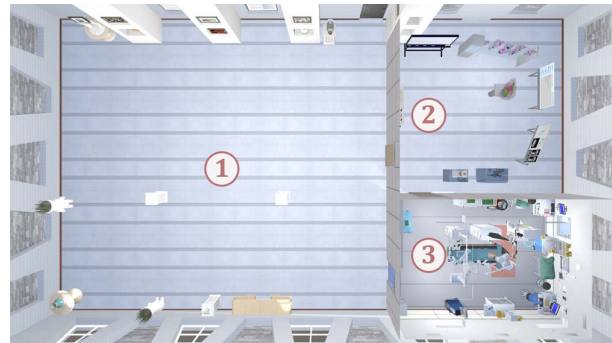


Figure 2: Overview of the environment, including a lobby (1), a room for surgical planning (2), and a virtual operating room for laparoscopic surgery training (3).



Figure 3: Patient datasets are presented on a virtual organ library (station 1) and can be interactively sorted according to their meta information, e.g., resection, tumor, and vessel type. The users can interactively investigate the individual case and they can use virtual whiteboard to write and make a discussion.

introduce themselves, hold a discussion, and explore the overview of the environment as well as the interactions listed on a virtual tablet, e.g., navigation technique.

3.2 Surgical Planning Room

For the surgical planning room, our aims are not only focused on virtual resection planning, but also on providing an entry-level surgical education of surgical planning through a problem-based learning approach. In a survey of virtual anatomy education systems [10], a high degree of visualization for organ models and their complex internal structures was considered to make it easier to memorize anatomical objects. Moreover, anatomy education in the collaborative VR environment could provide great potential benefits for education and training. It could enable collaborative learning by allowing students to participate and acquire knowledge effectively compared to the single-user systems.

3.2.1 Patient Cases

Within this room, users are allowed to choose, explore, and conduct virtual resection planning. A curated set of nine patient cases received from our clinical partner is presented in this environment. These patient cases represent the most frequently occurring tumors in the liver and a wide range of different anatomical variants and characteristics for resection planning. Each patient case includes 3D models, 2D medical image slices, medical and treatment information.

3.2.2 Four Surgical Planning Stations

There are four stations in the surgical planning room, including virtual organ library (station 1), information board (station 2), *DICOM*

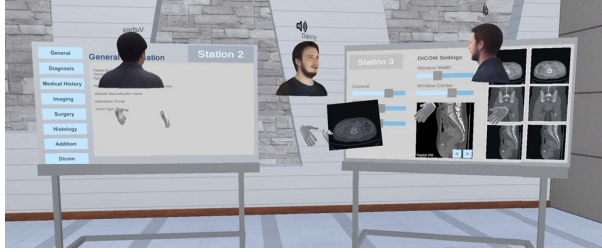


Figure 4: The users inspect the patient information, e.g., general information, diagnosis, and medical history (station 2) and explore medical image data (station 3).

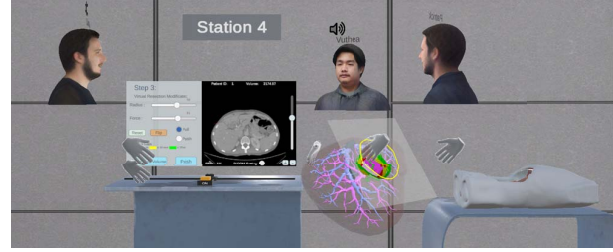


Figure 5: Surgical planning with virtual resection and risk map visualization (station 4).

board (station 3), and virtual resection planning (station 4).

- S1 *Virtual organ library* – The organ models of patient cases are arranged on the virtual compartments similar to the shelves in a library (see Fig. 3). Hence, an overview of all available cases for exploration and selection could be provided. In this station, the patient cases can be sorted according to different types of resection and their properties, e.g., tumor and vessel types. The users can select one of the cases from the virtual shelf to inspect further information and perform the planning. Additionally, there is a virtual whiteboard near the station which can be used for drawing illustrations and discussion.
- S2 *Information board* – After the users choose a patient case, they can inspect detailed information of the patient data on the information board (see Fig. 4). The information of the patient case is clustered into seven categories, including general information, diagnosis, medical history, imaging, surgery, histology, and additional information. With this station, the users can load additional medical image data to further explore on the DICOM board at the station 3.
- S3 *DICOM board* – A DICOM board allows the users to inspect the medical image data in detail (see also Fig. 4). Thus, they can use the basic slicing tools to change the image slices in different axes, e.g., coronal (back and front), sagittal (left and right), and axial (head and tail) planes. Furthermore, the adjustments of DICOM images are also provided to change the intensity and gray values of image data. It is also possible to select one image slice and hand it over to other users for further inspection and discussion.
- S4 *Virtual resection planning* – Within this station, the users can perform virtual resection planning and estimate resection volumes. First, they can place the liver model in a virtual mannequin to assess the position and orientation of the liver in the patient's abdomen. It is essential for planning, especially in laparoscopic surgery, because it provides further information regarding the tumor localization and the illustration for laparoscopic instruments' placement. Thereafter, the users can perform the virtual resection planning with the risk map visualization. Thus, they can draw incision lines, deform a resection surface, and estimate the resection volumes (see Fig. 5).

3.3 Virtual Operating Room

The main idea for the virtual operating room is to improve communication, train psychomotor skills, and provide as engaging a training environment as possible. The laparoscopic instruments (Simballs) are integrated to enable the training of these two roles: the surgeon performs the cutting and grasping and the assistant controls the laparoscopic camera. The interactions of both users are synchronized

in real time. During the laparoscopic cutting, critical events can occur, such as a bleeding vessel that needs to be clipped. Additionally, camera navigation and communication skills during these critical events between the surgeon and camera assistant for laparoscopic procedures can be trained.

3.3.1 Interprofessional team training

Teamwork and communication between surgeons and other professions during laparoscopic procedures are crucial for a fortunate surgical outcome. In this virtual operating room, we focus not only on the laparoscopic surgery training but also on the VR-based team training between the surgeons and anesthesiologists. Two training scenarios for interprofessional team training are enhanced and integrated [6] (see Fig. 6). These training scenarios contain critical events which are introduced to engage communication between the surgeon, the camera assistant, and the anesthesiologist, e.g., during undetected bleeding and insufficient muscle relaxant medication.

We use the *Pulse Physiology Engine* (Kitware, Inc., New York, USA) to achieve a realistic replication of the patient's vital signs. This could provide a better adaptation and flexibility for training scenarios, including surgical procedures and the administration of anesthetic medicine.



Figure 6: Interprofessional team for laparoscopic surgery training in the virtual operating room.

3.4 System Architecture

We use *Photon* networking (Exit Games GmbH, Hamburg, Germany) to handle network sessions and data synchronization. We set up a corresponding server to provide load-balancing services to handle voice communication, network sessions, and data synchronization between the clients. A synchronization approach proposed by Singhal et al. [28] was used to handle data synchronization. This approach is realized with the active replication states where the clients are responsible for sending periodic data, e.g., position, rotation, and animation. To provide stability of the connection and

prevent the server's bottleneck, the clients are responsible for individual graphics rendering and computation. A remote procedure calls (RPC) communication is used to distribute the data from one to other clients. We avoid sending a large amount of data, e.g., a batch of the array and unnecessary updates every frame. Instead, only virtual objects which are interacting and necessary objects are required for synchronization. Network conditions and client's machine capabilities could influence and affect the collaborative performance. The major factors, including delay, bandwidth, jitter, and packet loss are considered and optimized accordingly to provide good user experiences [29].

While we focus on VR-users, the proposed environment can be used with non-VR user using desktop-based PCs and conventional input devices. The non-VR users are connected as the spectators. Therefore, they can use mouse and keyboard to navigate in the environment.

3.4.1 Photo-realistic Avatars

We developed a small user management system that allows users to create their user accounts, upload their photos, and create a match-making room. Moreover, the system allows the users to choose their roles and scenarios before forming in the shared virtual environment.

Regarding photo-realistic avatars, we developed a layer and a user interface that allow the user to upload a single 2D photo and generate the 3D avatar that can be stored and used in the collaborative VR environment. With the support of *Avatar SDK* (itSeez3D, California, USA), the uploaded photo is generated to define the feature points of the 3D mesh and its corresponding texture.

3.5 Evaluation

The proposed environment was evaluated with domain experts. Six participants (two females) participated in the pilot study. Five of them (P1–P5) were general and visceral surgical residents with 8, 5, 3, 4, and 6 years of working experience, and one participant was a medical student (P6). All of them had previous VR experiences and were right-hand dominant. We followed a think-aloud protocol to collect general feedback. Furthermore, a semi-structured interview and a session for open questions were conducted after the experience of the system.

4 RESULTS AND DISCUSSION

In the following, we summarize the results of the participant's feedback into its relevant categories.

Lobby P3 and P5 stated that the lobby could be useful to allow the users to find each other after joining the environment and to explore the navigation technique. P3 commented that using virtual tablet was helpful to change the configuration, e.g., navigation techniques between the teleportation and steering techniques. However, they had some difficulties keeping the virtual tablet stays active. Thus, they would prefer to press the controller's trigger button once to activate and press it again to deactivate, rather than press and hold it.

Usefulness for surgical planning and training All participants were positive and confirmed the potential benefits of this proposed environment. For the surgical planning room, four stations were well integrated. It was essential to provide the planning and training that ranges from case selection to operating room training. P1 stated that the current training for laparoscopic procedure in their department often uses the commercial laparoscopic simulator with the standard 3D model. Thus, it could be advantageous for planning and training with real patient cases. Moreover, they expressed that linking between the selected patient case from the planning room to be used for training in the operating room is beneficial to train and understand the surgical procedures. P2 stated that the proposed environment could be extremely useful for planning complex patient

cases and training surgical planning procedures for laparoscopic surgery. They found that the virtual mannequin at station 4 was advantageous in providing the planning for laparoscopic surgery compared to the previous planning software. In addition, P2 commented that the proposed environment could be applied for their surgical training routine, especially for junior surgeons. Regrading the training for medical students, P4 stated that allowing the students to join the environment as spectators without using VR headsets was a useful function. Therefore, in some cases, medical experts can use this system to conduct the surgical planning so medical students can also join as spectators to learn and understand the procedures.

Photo-realistic avatars P1 expressed that this feature was a good addition to the proposed collaborative VR environment. They stated that allowing the users to have their own avatars generated from their photos could be advantageous in providing an engaging environment. They could use the user management system to create user accounts and generate avatars. The blinking of the generated avatar was randomized and the lip movement was animated according to the voice communication. It could provide more realistic avatars if the system detects the eye and lip movements of the users by using additional sensors. Nonetheless, they commented that it was adequate for the current state because they could notify with an additional chat icon appearing on top of the avatar if one user engages the voice communication.

Overall, the proposed collaborative VR environment was evaluated as an essential tool for planning and training the laparoscopic procedures. The results of the pilot study was followed by the think-aloud protocol and a semi-structured interview. The participants confirmed its potential benefits and clinical applicability. In the surgical planning room, they found that the stations were well integrated. In addition to the virtual resection planning, the virtual mannequin was assessed as an important tool to evaluate the resection planning for laparoscopic setup. Besides the planning stations, P5 and P6 commented that the interactions with the virtual whiteboard were intuitive in drawing the illustration for further team discussion. In the virtual operating room, the connection of the patient case that was selected in the planning room was assessed as beneficial.

5 CONCLUSION

We have introduced a collaborative VR environment of a virtual hospital to improve different aspects of surgical planning and training. In the surgical planning room, we integrated four planning stations ranging from patient case selection to virtual resection planning. For the virtual operating room, we presented a training system which can simulate different laparoscopic surgery procedures for connected users in different roles. Hence, they can practice, communicate, and improve surgical skills, respectively. The integration of the work with interprofessional team training in laparoscopic surgery was also presented, which allows the surgeons and anesthesiologists to practice their communication and teamwork in different critical events. In addition, we presented the development of photo-realistic avatars and interaction techniques to provide an engaging training environment.

The collection of patient cases can support surgeons during the decision process. Therefore, they can sort and evaluate the cases, e.g., tumor and vessel types. After selecting one case for planning, they can explore the 3D models and 2D image data as well as patient-related information. The planning procedure can further be supported by virtual resection planning with risk map visualization around the tumor. The results of resection and remaining liver volumes can be evaluated. Besides the planning with the 3D model in mid-air, they can assess the position and identify the tumor localization of the patient case inside the virtual mannequin. The selected patient case is also used in the virtual operating room to train with the laparoscopic simulation. Here, it can be determined if the planned virtual resection can be performed in the laparoscopic

surgery. Additionally, training scenarios and team collaboration with the anesthesiologists can be performed to support the operating room training. Future work aims to conduct a full clinical evaluation and investigate other training departments. Furthermore, group navigation and spatial audio are the important components to provide an effective collaboration in the virtual environment.

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